

A One-Dimensional Integral Representation for Diagonal Geode Coefficients, with the values of $G(1000, \dots, 1000)$ in dimensions 5 through 10

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Abstract

Amdeberhan, Kauers and Zeilberger offered a donation to the OEIS for the computation of the five-dimensional Geode coefficient $G[1000, 1000, 1000, 1000, 1000]$; Rubine, who computed the four-dimensional value in 35.9 hours, estimated the five-dimensional case at roughly a thousand times that cost and wrote that “new innovations are required.” We derive a representation of the diagonal Geode coefficient $G[M^{(D)}]$ as a single coefficient extraction against a one-dimensional integral, yielding an algorithm using $O(M^2)$ arithmetic operations and $O(M)$ memory per prime in a multi-modular scheme. The five-dimensional value (8367 digits) then takes minutes on commodity hardware. As validation, the same program reproduces Rubine’s published $G[1000^{(4)}]$ exactly in under three minutes, and we verify every reconstructed value against fresh primes not used in the reconstruction. We report $G[1000^{(D)}]$ for $D = 5, 6, \dots, 10$, all computed for the first time.

1 The challenge

Let $S = 1 + \sum_{k \geq 2} t_k S^k$ be the hyper-Catalan generating series of Wildberger–Rubine, restricted to D variables $t_2, \dots, t_{D+1}$, and let G be the Geode, $S - 1 = (t_2 + \dots + t_{D+1})G$. Write $G[M^{(D)}]$ for the coefficient of $(t_2 \cdots t_{D+1})^M$ in G . The hyper-Catalan coefficients have the classical closed form (Erdélyi–Etherington)

$$C[\mathbf{m}] = \frac{K!}{(1 + K - N)! \prod_k m_k!}, \quad K = \sum_k k m_k, \quad N = \sum_k m_k.$$

Expanding $1/(t_2 + \dots + t_{D+1})$ geometrically in t_2 gives, with $d = D - 1$,

$$G[M^{(D)}] = \sum_{\alpha \in \{0, \dots, M\}^d} (-1)^{|\alpha|} \binom{|\alpha|}{\alpha} C[M+1+|\alpha|, M-\alpha_1, \dots, M-\alpha_d], \quad (1)$$

a sum with $(M+1)^d$ terms (10^{12} at $M = 1000$, $D = 5$): the wall described by Rubine.

2 The collapse

Three elementary structural facts dismantle (??). Write $W = \sum_{j=1}^d j \alpha_j$, $b_K = M \sum_{k=2}^{D+1} k + 2$, $b_N = M \sum_{k=1}^D k + 1$.

Lemma 1. For every α in (??), the vector $\mathbf{m}'$ appearing in $C[\cdot]$ satisfies $N' = \sum_k m'_k = DM + 1$ (independent of α) and $K' = \sum_k k m'_k = b_K - W$.

Proof. $N' = (M+1+|\alpha|) + \sum_j (M - \alpha_j) = DM + 1$. For K' , the shift of m_2 contributes $2|\alpha|$ while the decrements remove $\sum_j (j+2)\alpha_j = W + 2|\alpha|$. $\square$

Consequently $K'!/(1+K'-N)! = (K')_{DM} = (DM)! \binom{K'}{DM}$, a polynomial in W , and $\binom{K'}{DM} = [z^{DM}](1+z)^{b_K-W}$.

Lemma 2. With $T[i, W] = \sum_{|\alpha|=i, W(\alpha)=W} \prod_j \binom{M}{\alpha_j}$ one has $\sum_{i,W} T[i, W] u^i y^W = \prod_{j=1}^d (1 + u y^j)^M$.

Proof. The binomial theorem in each factor. $\square$

Lemma 3. $\frac{i!}{(M+1+i)!} = \frac{1}{M!} \int_0^1 x^i (1-x)^M dx$.

Proof. The Beta integral. $\square$

Substituting Lemma ?? into (??), using $1/(\alpha_j!(M - \alpha_j)!) = \binom{M}{\alpha_j}/M!$, then evaluating the α -sum by Lemma ?? at $u = -x$ via Lemma ??, gives:

Theorem 1. With $d = D - 1$, $s_d = d(d+1)/2$, $E = (\sum_{k=2}^{D+1} k - s_d)M + 2$ and $h(x, z) = (1-x) \prod_{k=1}^d ((1+z)^k - x)$,

$$G[M^{(D)}] = \frac{(DM)!}{(M!)^D} [z^{DM}] (1+z)^E \int_0^1 h(x, z)^M dx.$$

3 Algorithm

Modulo a prime p , the integral of the polynomial $h(\cdot, z)^M$ (of x -degree DM) is evaluated exactly by interpolatory quadrature at $DM + 1$ nodes x_j (avoiding $x = 1$); at equispaced nodes the Lagrange denominators are $(-1)^{n-1-j} j!(n-1-j)!$. For each node, $g_j(z) = h(x_j, z)$ has degree s_d , and the coefficients of g_j^M satisfy the order- s_d recurrence obtained from $g P' = M g' P$, costing $O(s_d)$ operations per coefficient; they are contracted on the fly against $\binom{E}{DM-r}$ and the quadrature weights. The cost is $O(D^2 s_d M^2)$ arithmetic operations and $O(DM)$ memory per prime; $\approx 10^3 - 1.6 \times 10^3$ 30-bit primes reconstruct the values below by the CRT. A proven bound $|G[M^{(D)}]| \leq \frac{b_K!}{(1+b_N)!(M!)^d} \sum_i \frac{i! \binom{dM}{i}}{(M+1+i)!}$ (from $\sum_W T[i, W] = \binom{dM}{i}$) and the monotonicity of $W \mapsto (b_K - W)_{DM}$ sizes the prime count; we found empirically that the naïve leading-term estimate is *not* an upper bound and must not be used for this purpose.

4 Validation

(i) a five-stage machine verification (closed form vs. the defining series; (??) vs. exact polynomial division and vs. an independent recursion; each lemma separately; Theorem ?? vs. exact values; production code vs. exact values for $D = 4, \dots, 10$ at small M); (ii) exact reproduction of Rubine's published $G[1000^{(4)}]$ (6036 digits; our wall time 168s on a 13.6-core allocation against his reported 35.9 h), after matching his printed $H(1), \dots, H(6)$ exactly; (iii) every reconstructed value verified against fresh primes not used in its reconstruction. Code, residues and full decimal values: <https://github.com/jaideepsaipadhi/5d-geode-challenge>.

5 Results

D	digits	first 12 digits	last 12 digits
4	6036	140604899259	407713629184000 (Rubine ✓)
5	8367	365748434524	213507850240000
6	10866	354720604493	172747182080000
7	13506	203334720825	405302272000000
8	16267	256748206726	425216000000
9	19135	132148278181	033280000000
10	22097	772529711009	848000000000

Remarks. The representation is uniform in D ; nothing changes structurally as the dimension grows except s_d . The same integral representation applies off the diagonal with $\prod_k$ carrying distinct exponents, and may be of independent interest for the asymptotics of diagonal Geode coefficients.

Acknowledgements

[TODO]

References

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